

MADHAV WADHWANA

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EXPERIENCE

AI Engineer | Python Developer

Softrefine Technology Pvt. Ltd.

December 2025 - Present

Ahmedabad, Gujarat

- Developed AI-powered backend services and Flask APIs for content-generation platforms.
- Integrated LLM-powered workflows and structured output validation pipelines for reliable AI-generated content.
- Designed JSON schema validation layers to ensure consistency between AI responses, templates, and frontend rendering.
- Built template-driven rendering systems using HTML/Jinja with dynamic asset injection.
- Debugged and optimized production AI-enabled backend services for scalability, performance, and maintainability.

React Native Developer (Part-time)

Timeverse Ltd.

July 2025 – December 2025

Ahmedabad, Gujarat

- Developing and maintaining the entire application stack, covering both frontend (React Native) and backend services.
- Built scalable mobile experiences with smooth navigation, animations, and robust state management.
- Designed and implemented reusable React components, optimized performance, and ensured pixel-perfect UI across devices.
- Developed and integrated RESTful APIs, managed database operations, and set up deployment pipelines for production-ready systems.
- Collaborated with cross-functional teams to deliver end-to-end features from concept to deployment.

Product Development Intern

VisiLean Pvt. Ltd.

July 2024 - October 2024

Ahmedabad, Gujarat

- Leveraged GitHub *GraphQL* API for data ingestion and ETL processing, generating JSON-based analytical reports to enhance repo intelligence.
- Implemented a *Cesium JS*-powered 3D geospatial visualization module, augmenting UI/UX interactivity for geospatial analytics.
- Refactored and optimized Power BI dashboards, streamlining data pipelines and enhancing BI visualizations for data-driven decision-making.
- Synergized with cross-functional Agile squads throughout the *SDLC*, driving seamless *CI/CD* workflows from design inception to production rollout.

PATENTS

Plantation and Habitat Control Device for Vegetative Propagated Crops

Patent No: 2023210141007 A | Filed Jul 16, 2025

- A cuboidal device equipped with motorized wheels for maneuvering across farm fields.
- Includes a storage chamber, screw conveyor, processing chamber, and motorized cutter to prepare crop segments for sowing.
- Features a soil quality assessment module that provides real-time ground condition analysis.
- Equipped with fertilizer and water storage systems, regulated through valves and pipes, and a robotic arm with a digging unit for automated sowing.

Mobile Device for Rejuvenating Diabetic Patients

Patent No: 202521068003 | Filed Jul 16, 2025

- A mobile device mounted on wheels with a data input module to capture user profile, medical prescriptions, and desired health goals.
- Incorporates an image sensing module for user identification and a wearable band to track real-time vital parameters.
- Features a food preparation module that generates customized diets by comparing prescriptions, vitals, and goals with a prestored database.
- Includes a blood glucose monitor, smart glasses for guided exercises, and robotic arms to physically support the user during workouts.

SKILLS

Languages	C/C++, Python, Java(Beginner), HTML, SQL
Libraries/Frameworks:	PyTorch, TensorFlow, Git, Docker, OpenCV, Flask, Django, Spring Boot(Beginner)
Technologies:	Visual Studio, Microsoft 365, AWS (EC2, SageMaker, etc), Azure, HuggingFace
Databases:	PostgreSQL, MongoDB
Frontend:	React, React Native, JavaScript, TypeScript, Tailwind CSS, Redux
Generative AI:	LLM Applications, Prompt Engineering, Structured Output Generation, Hugging Face, API-
Backend & APIs:	based LLM Integrations Node.js, Flask, Django, REST APIs

EDUCATION

Marwadi University, India	July 2021 - May 2025
Bachelor's in Computer Engineering, Specialization in Artificial Intelligence	
Relevant Coursework: Machine Learning, Deep Learning, Data Science, Natural Language Processing (NLP), Representation Learning, Large Language Models (LLMs)	

PROJECTS

3D Globe Integration	Github
• Developed and integrated a 3D globe feature into a web application using Cesium JS. Enhanced user interaction with geographical data by enabling interactive geographical visualizations. The globe was designed to be fully responsive and accessible across different devices.	
Credit Card Input	Github
• Built and customized a React Native credit card input component with real-time validation and smooth animations to enhance secure mobile payment experiences.	
Billing System using Java Swing & JDBC	Github
• Engineered a Billing System with Java Swing and JDBC, enabling users to manage transactions, generate invoices, and maintain a sales database efficiently.	
LLM-Powered AI Assistant	Github
• Developed a cross-platform React Native app integrated with DeepSeek API to deliver AI-generated responses in real-time with streaming. Implemented low-latency, context-aware conversational features, multi-platform support, and a customizable UI using React Native Reanimated, React Context API, and Tailwind CSS.	
Attendance Tracker using JSP, JDBC, HTML & CSS	Github
• Developed an Attendance Tracker using JSP, JDBC, HTML, and CSS, facilitating simplified tracking and management of attendance records for educational or organizational purposes.	